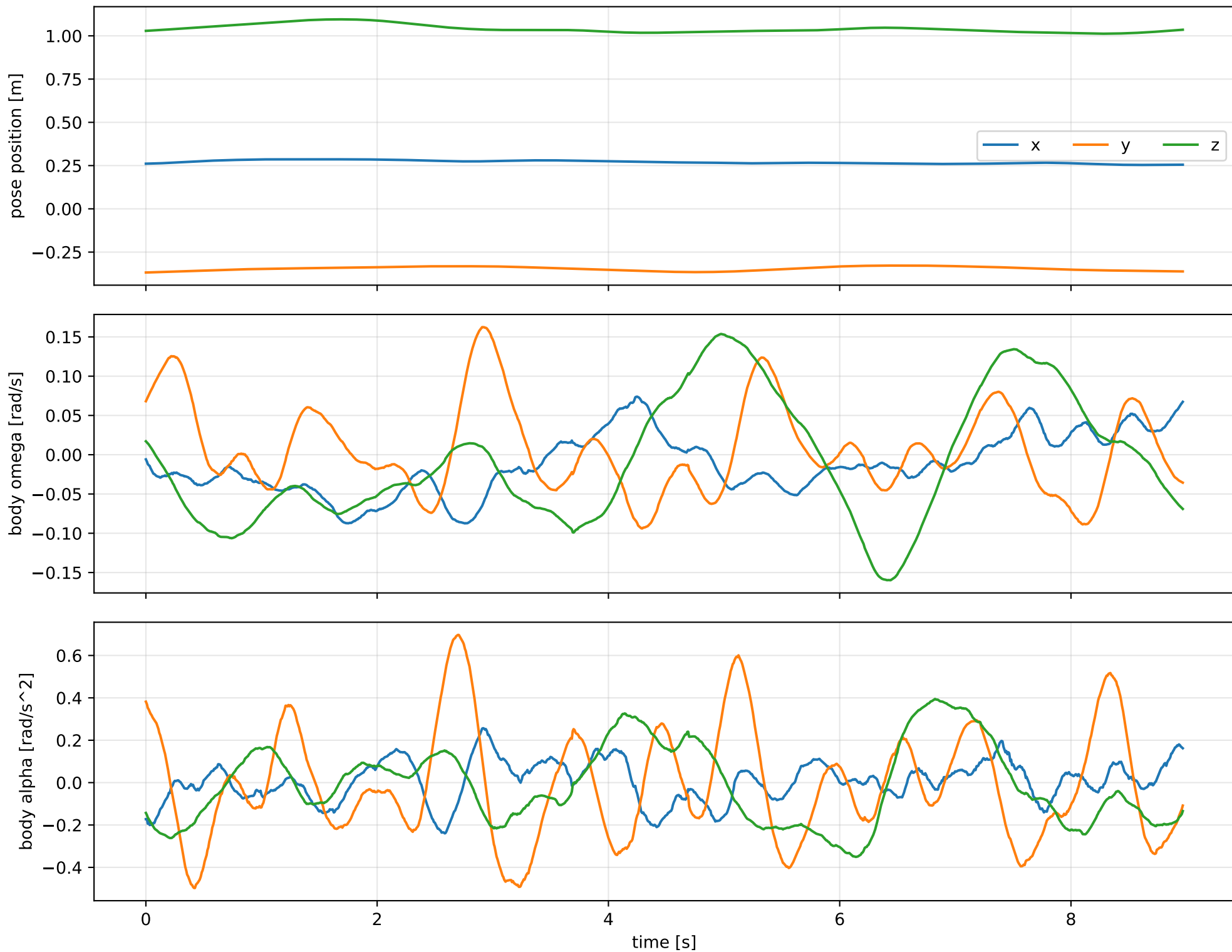
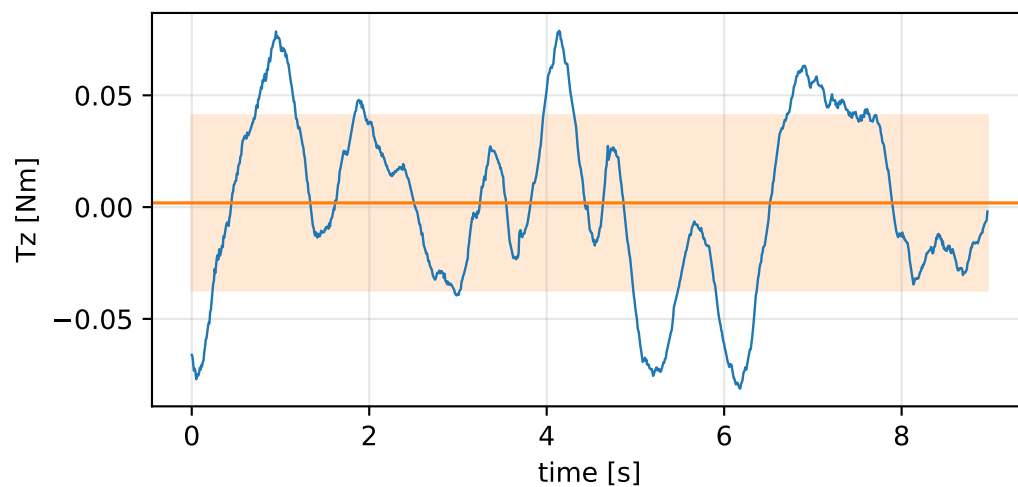
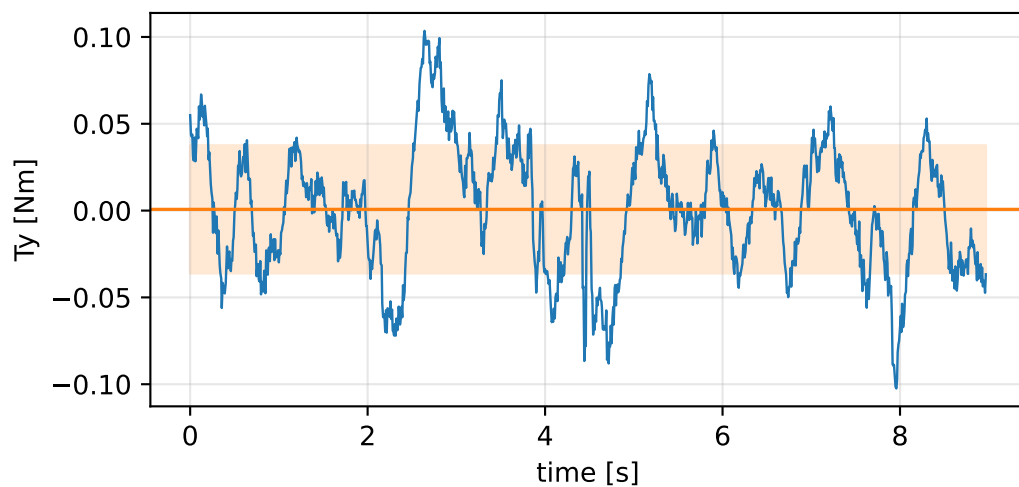
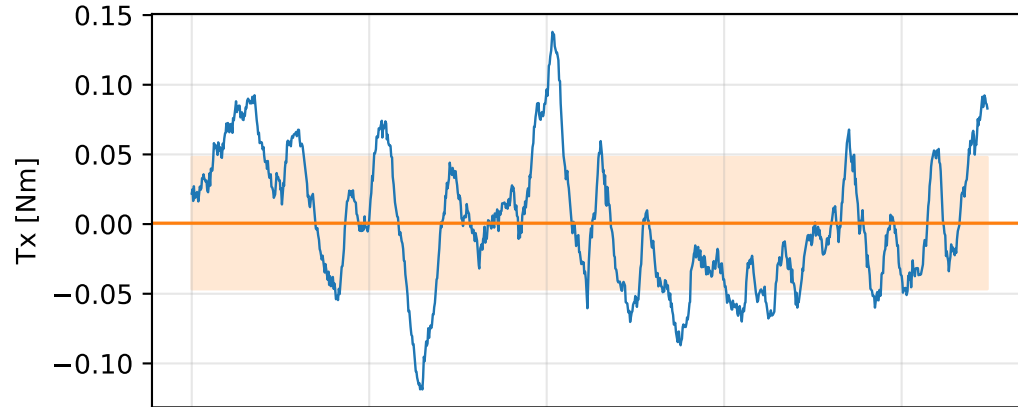
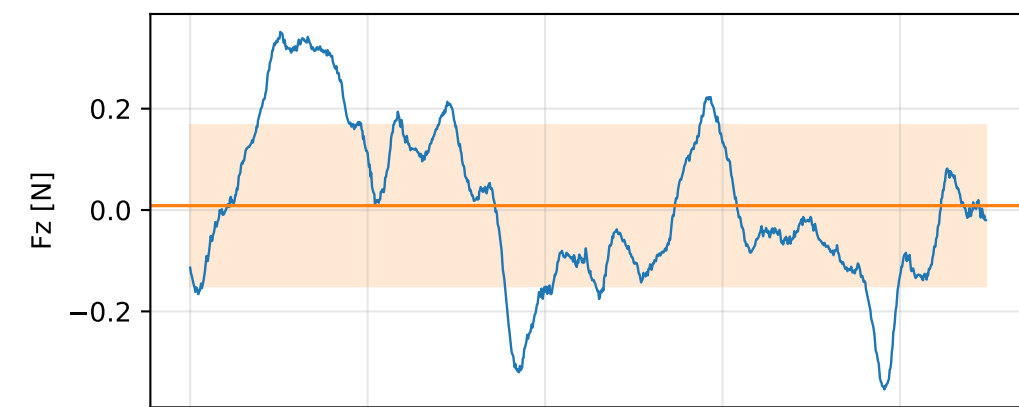
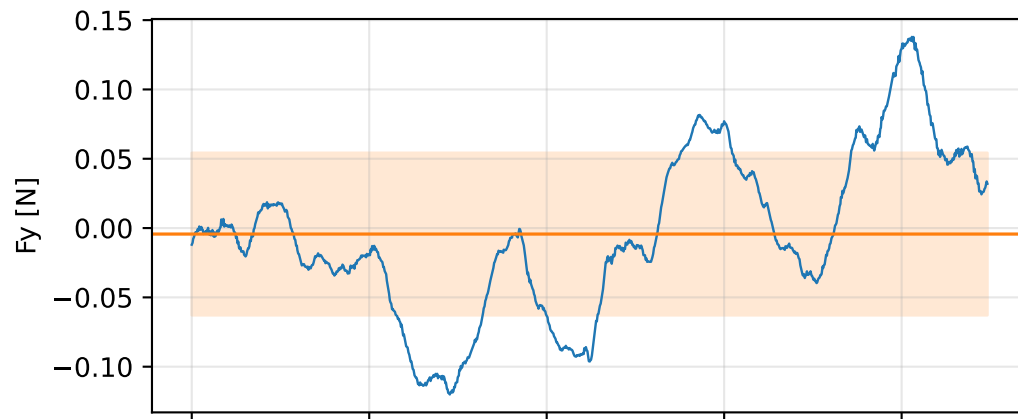
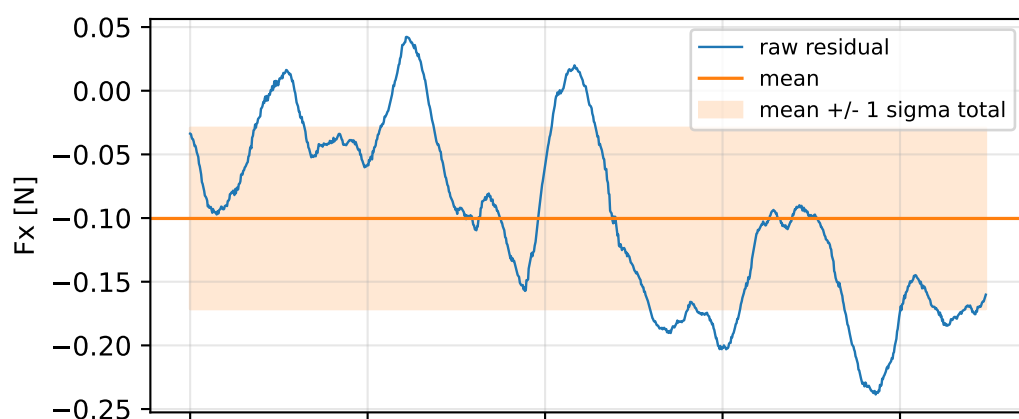


inertia_over_mass_yz_nominal: SG trajectory and derived motion



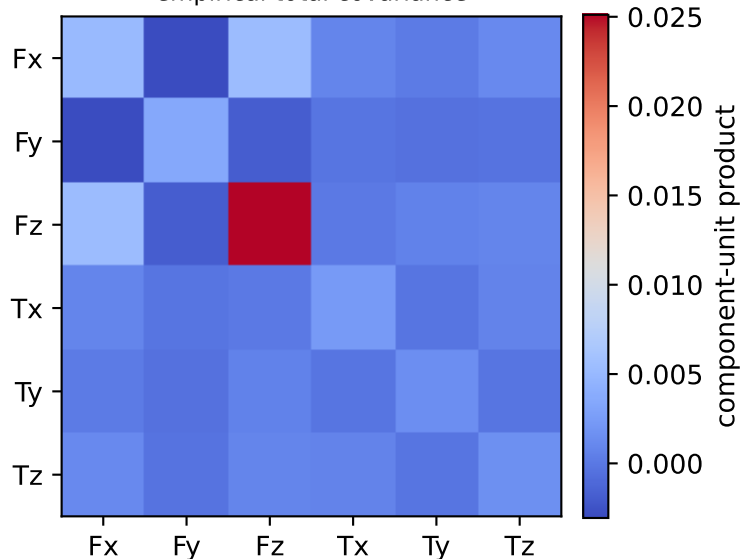
inertia_over_mass_yz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



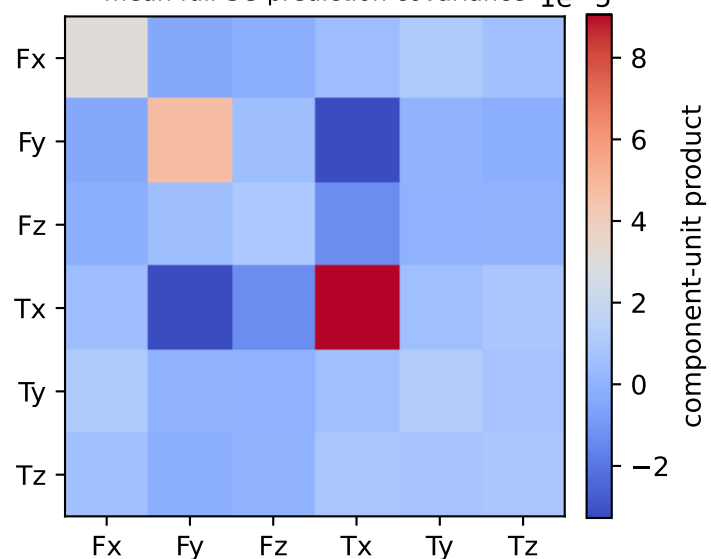
vector RMS about zero: force 0.20909 N, torque 0.071551 Nm; component std about mean: force [0.0711 0.0586 0.1585], torque [0.0473 0.0368 0.0391]

inertia_over_mass_yz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.003 0.006 0.027]

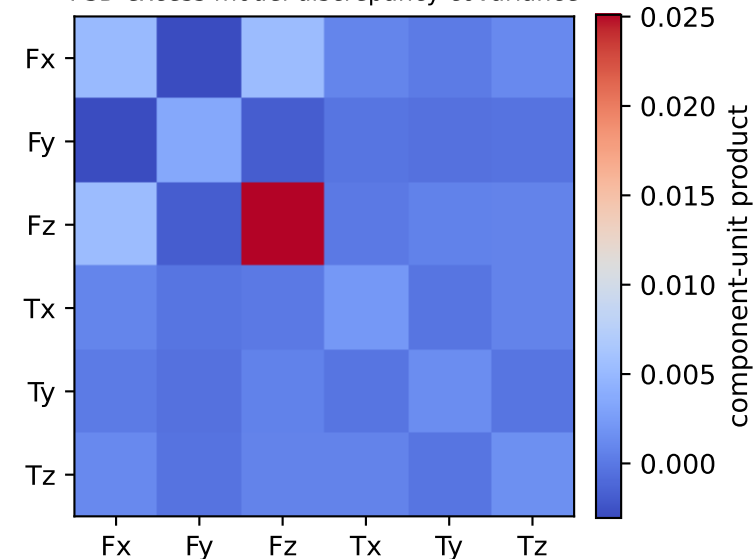
empirical total covariance



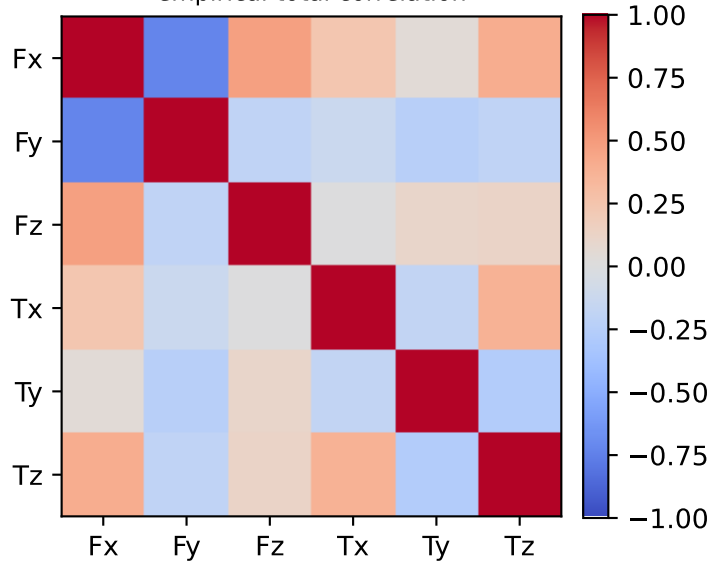
mean full-SG prediction covariance $1e-5$



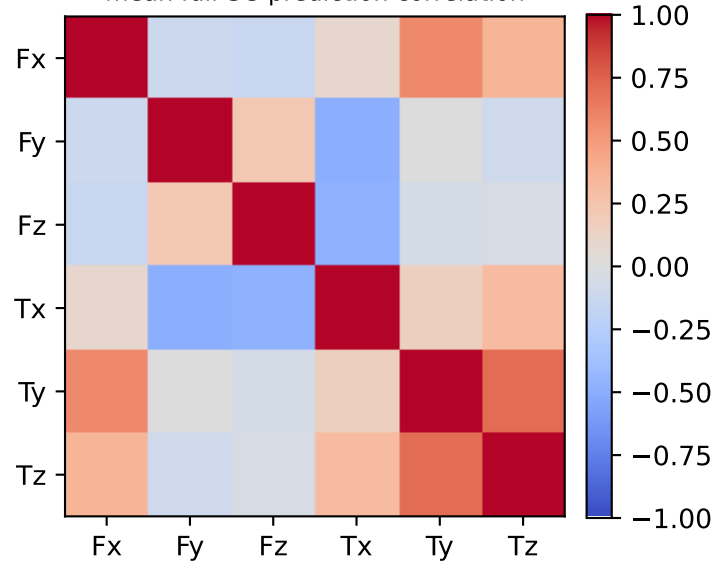
PSD excess model discrepancy covariance



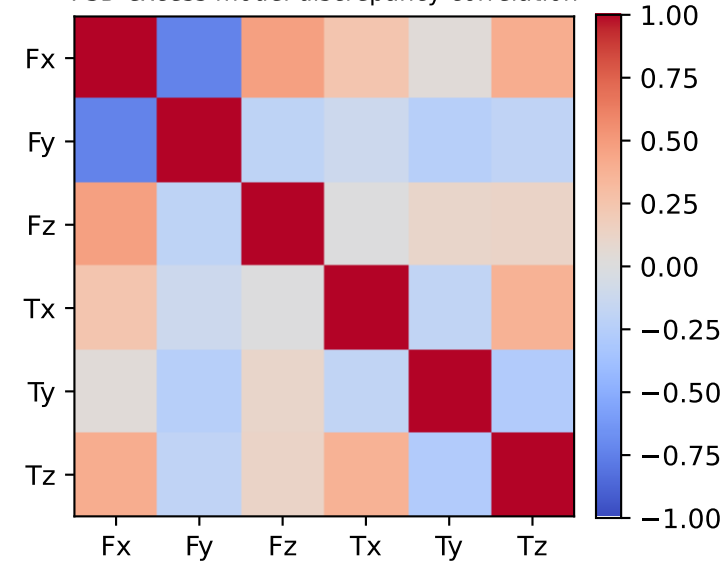
empirical total correlation



mean full-SG prediction correlation

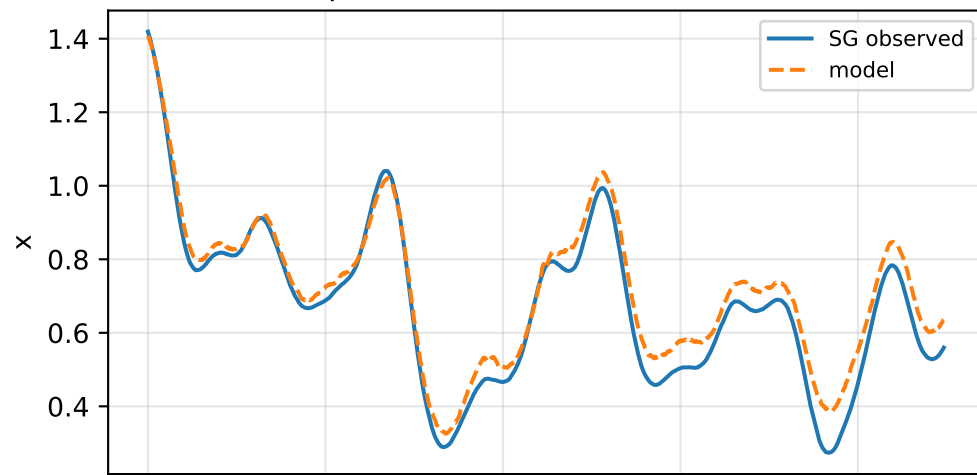


PSD excess model discrepancy correlation

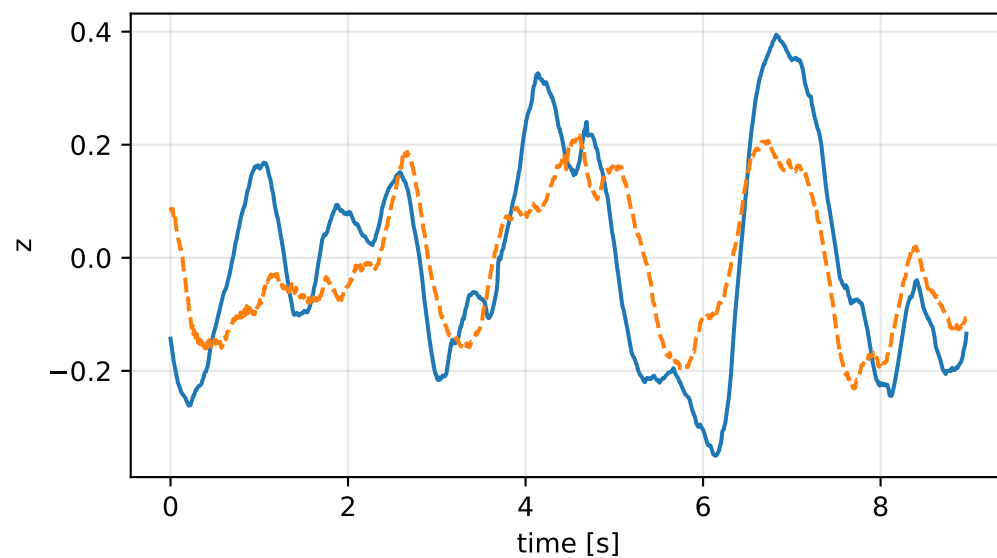
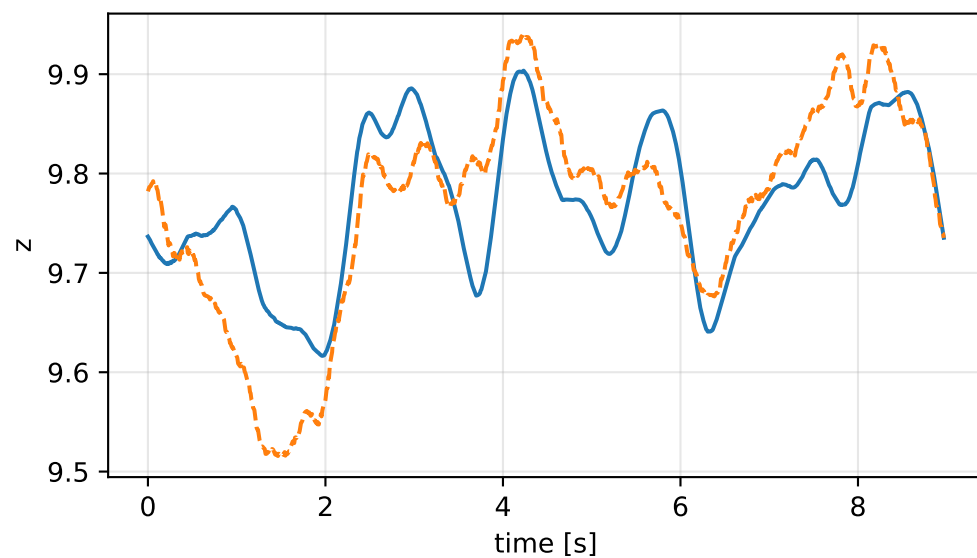
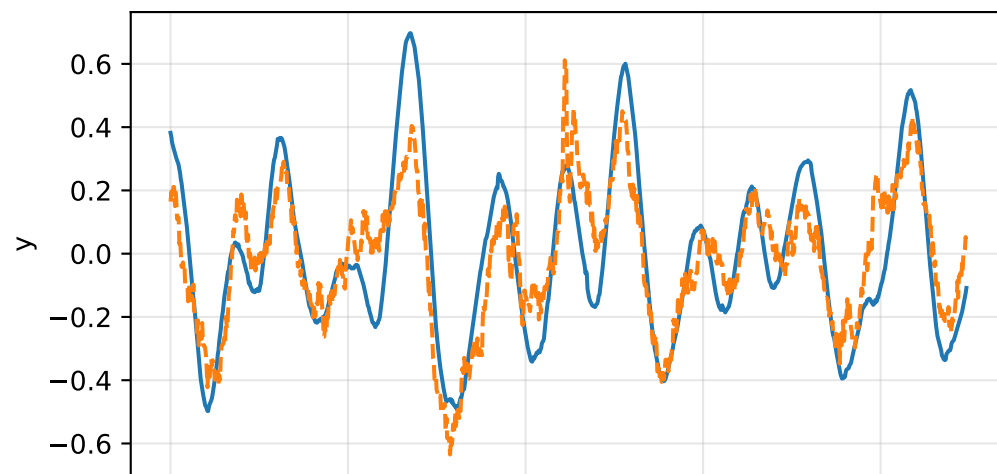
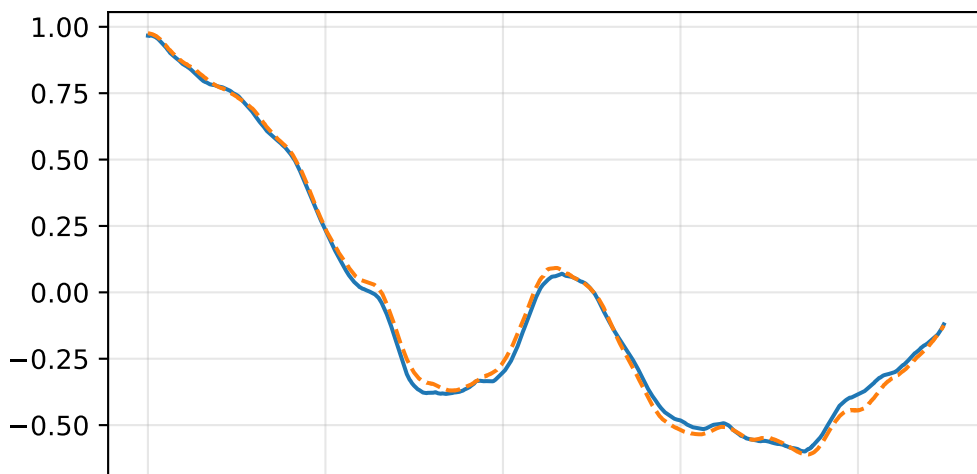
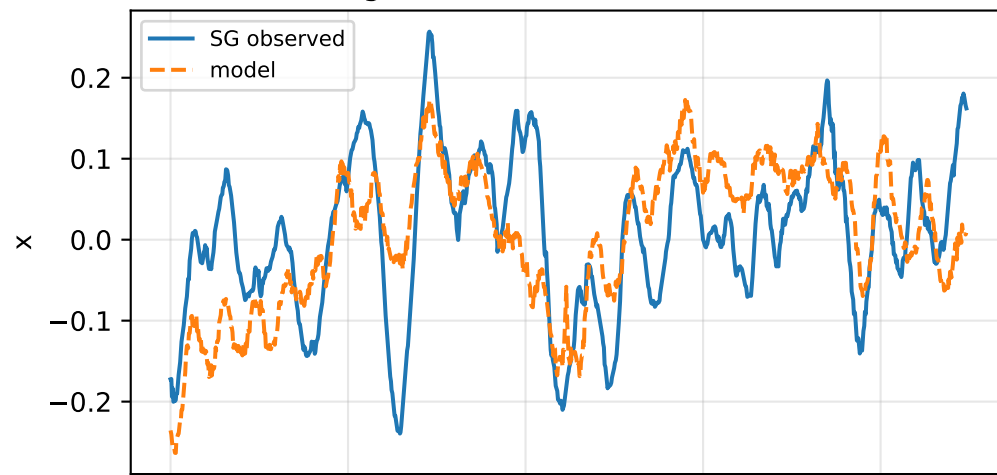


inertia_over_mass_yz_nominal: acceleration residual objective

specific acceleration [m/s²]

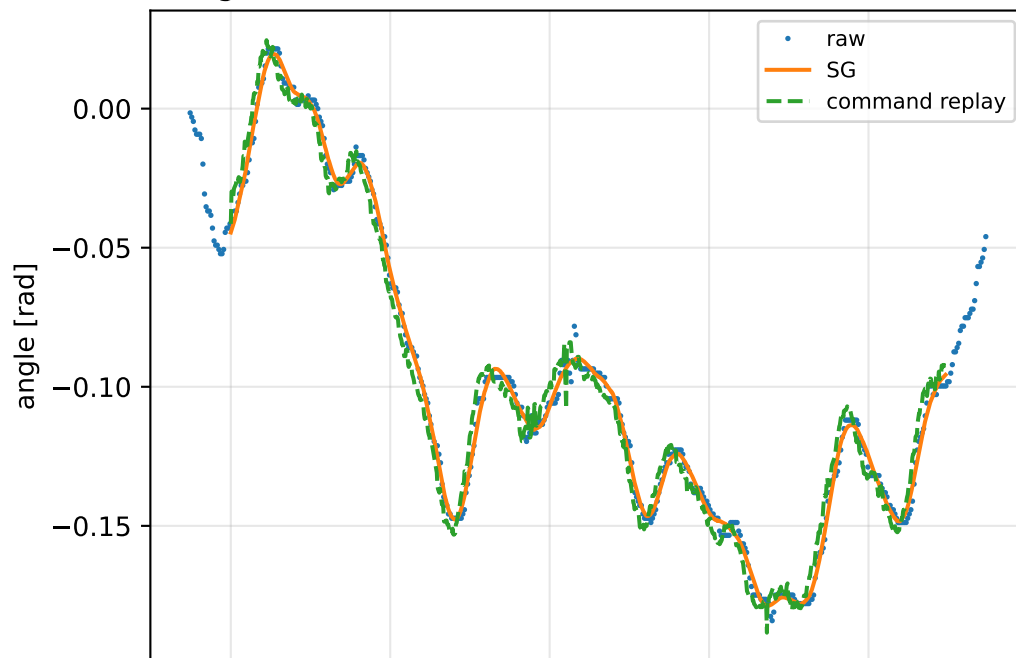


angular acceleration [rad/s²]

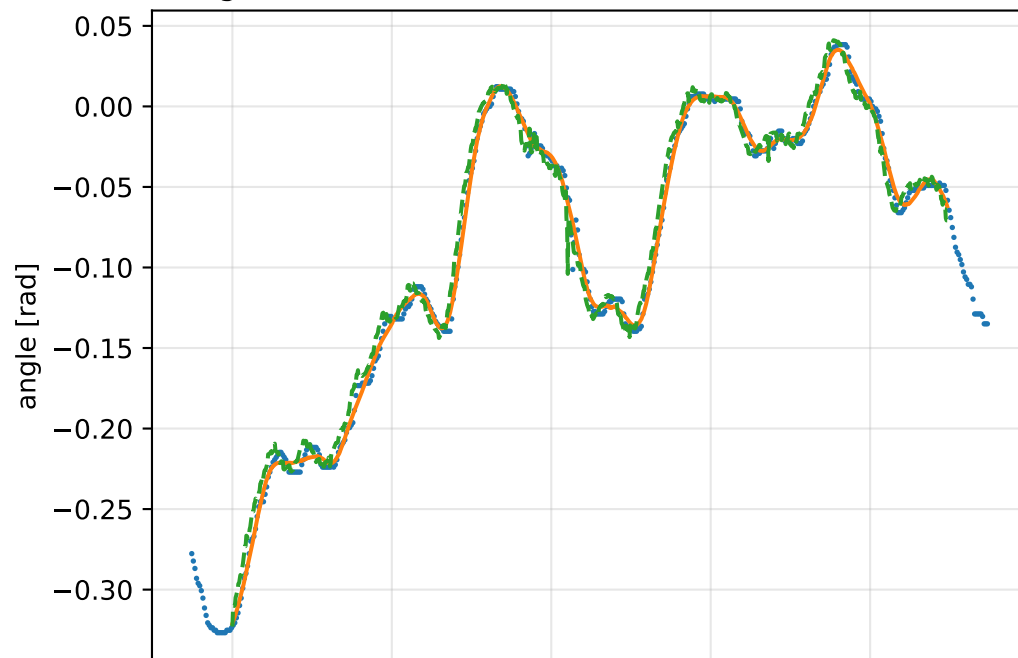


inertia_over_mass_yz_nominal: gimbal measurement audit (objective=measured_sg)

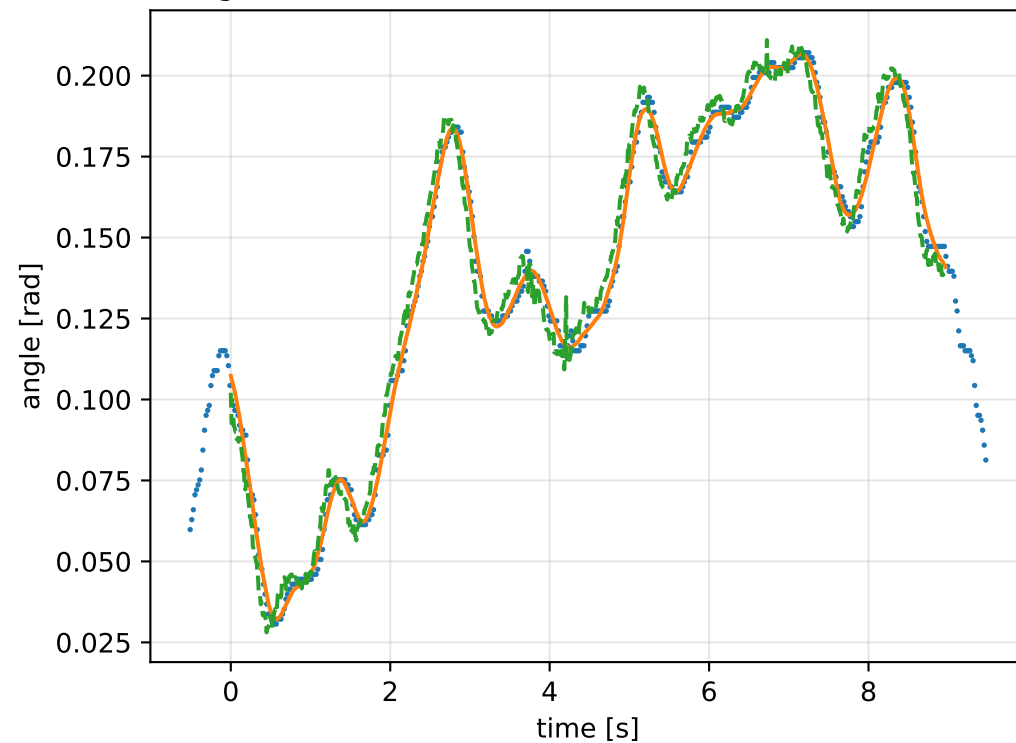
gimbal 1: RMSE=0.006345, max=0.01494 rad



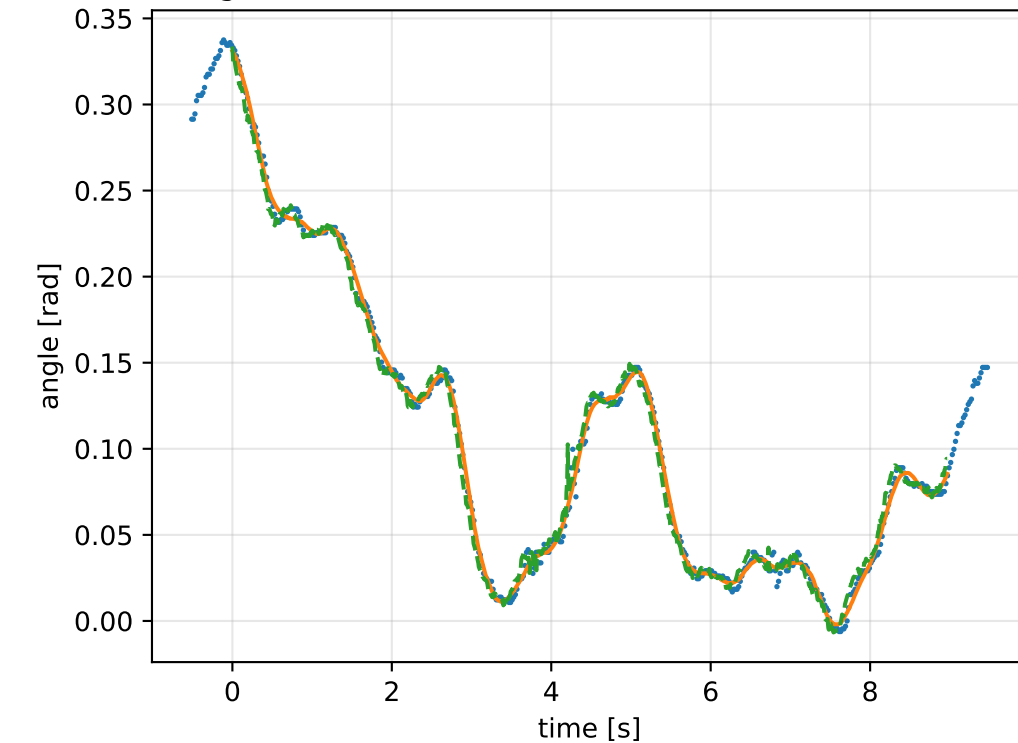
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

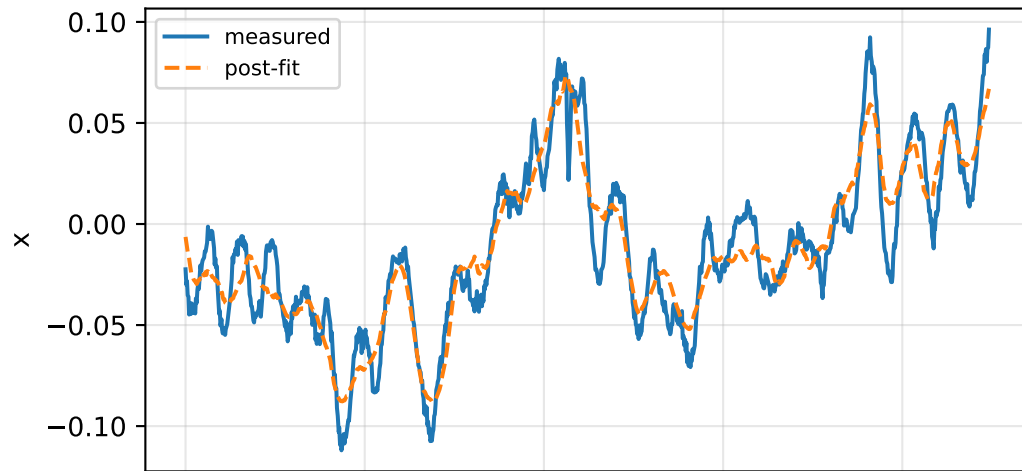


gimbal 4: RMSE=0.006321, max=0.03642 rad

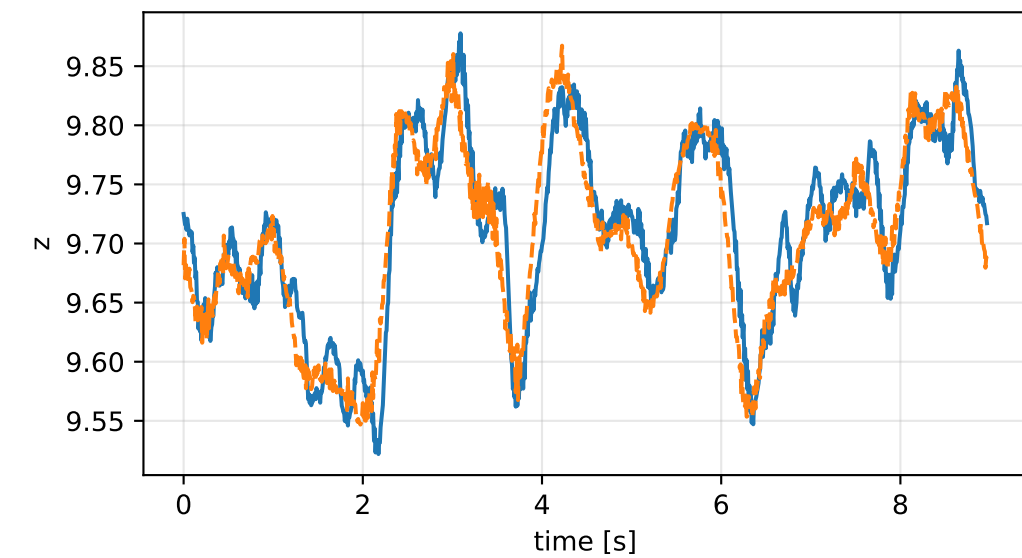
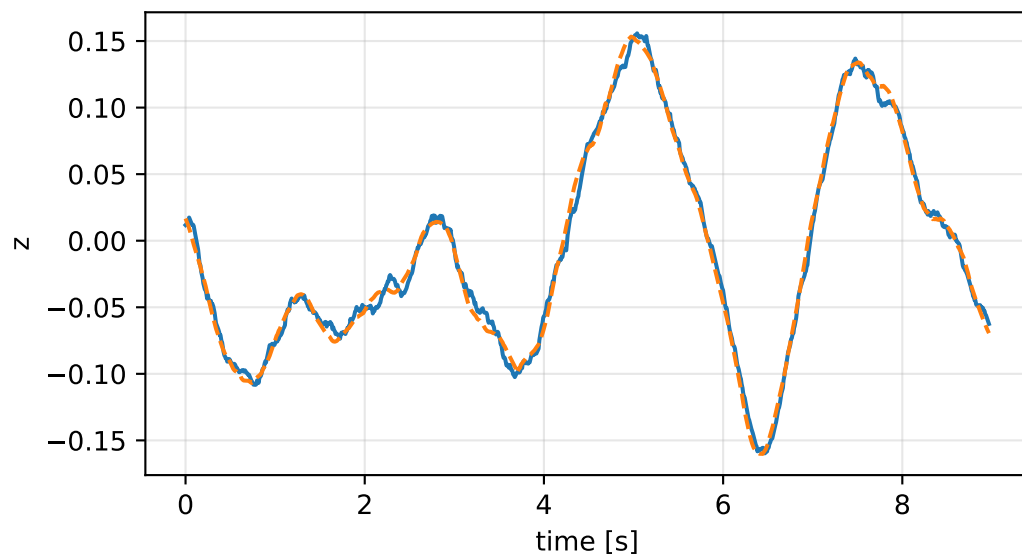
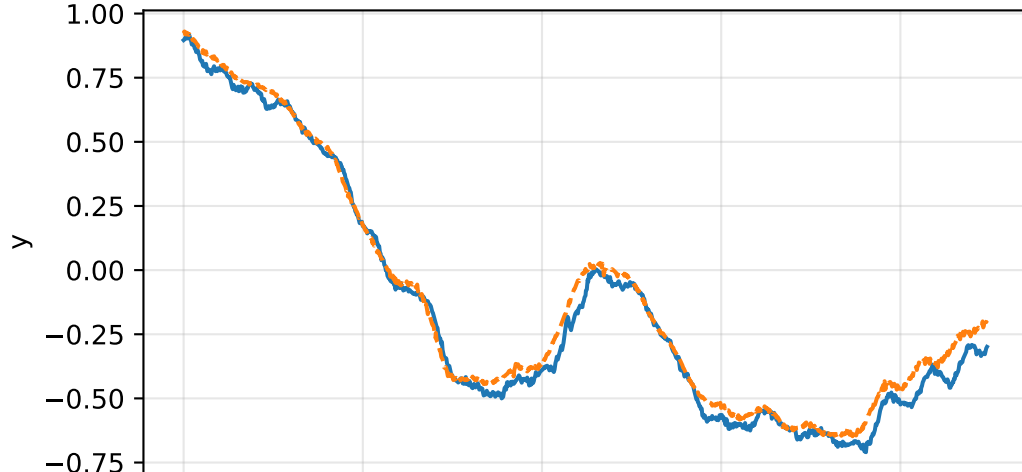
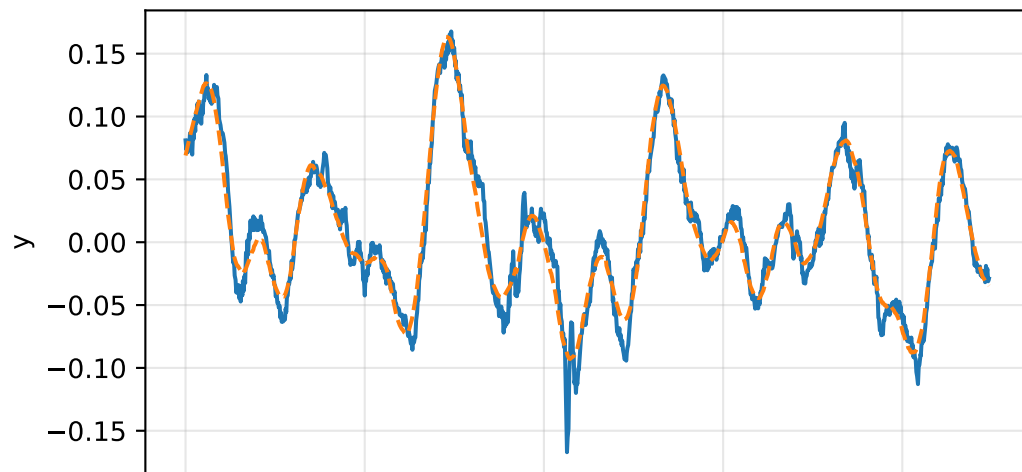
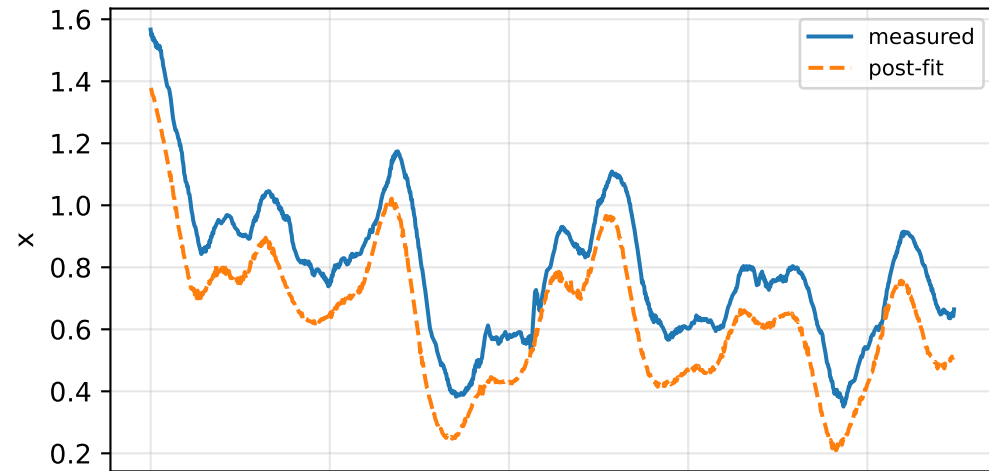


inertia_over_mass_yz_nominal: IMU comparison (diagnostic only)

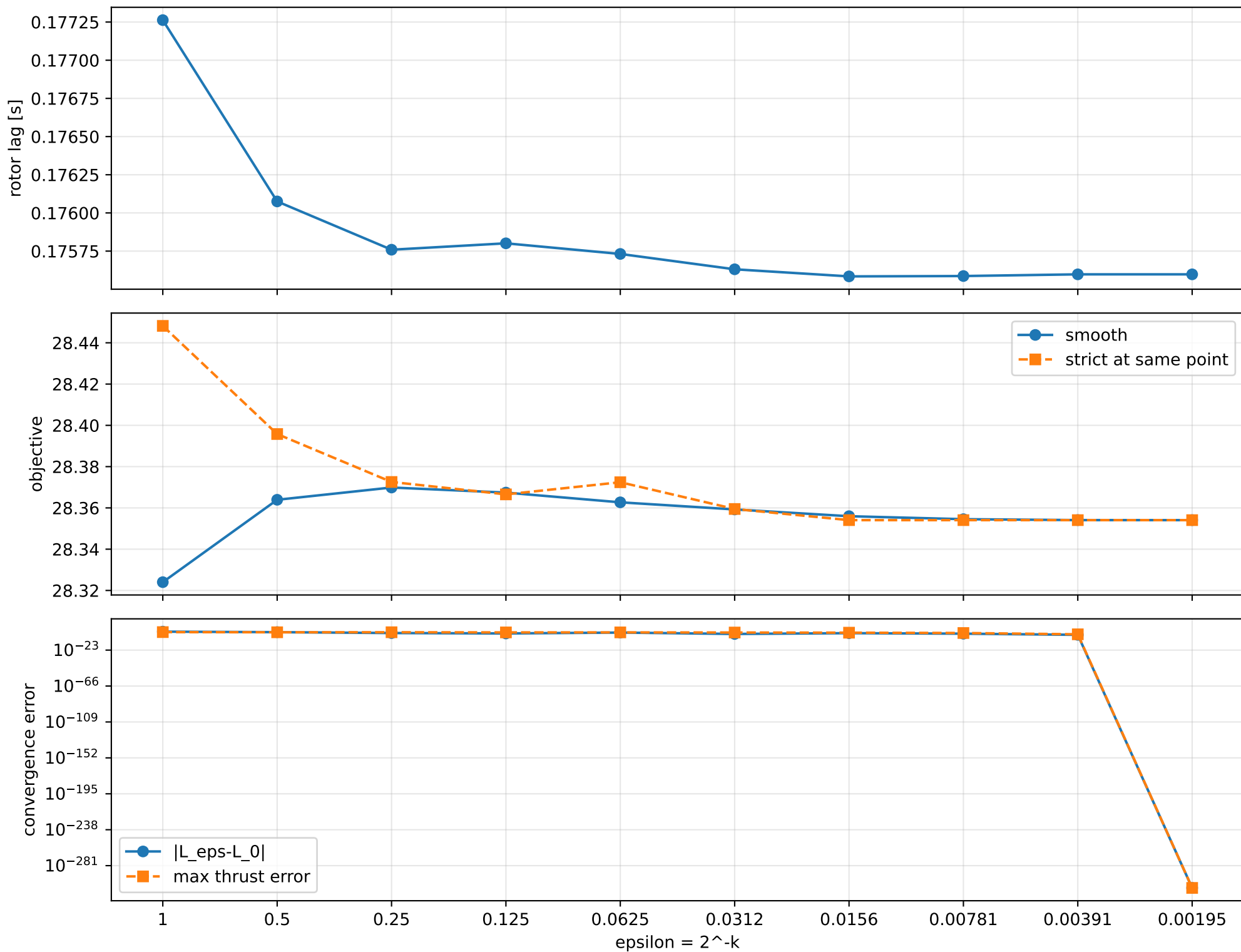
gyro [rad/s]



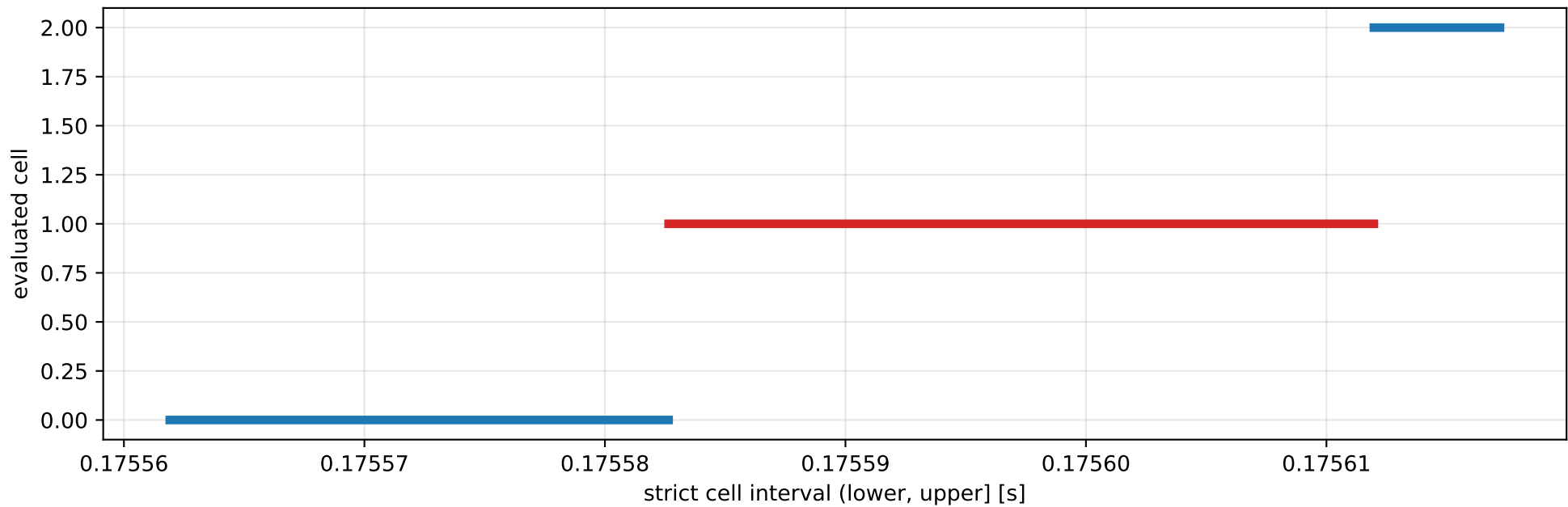
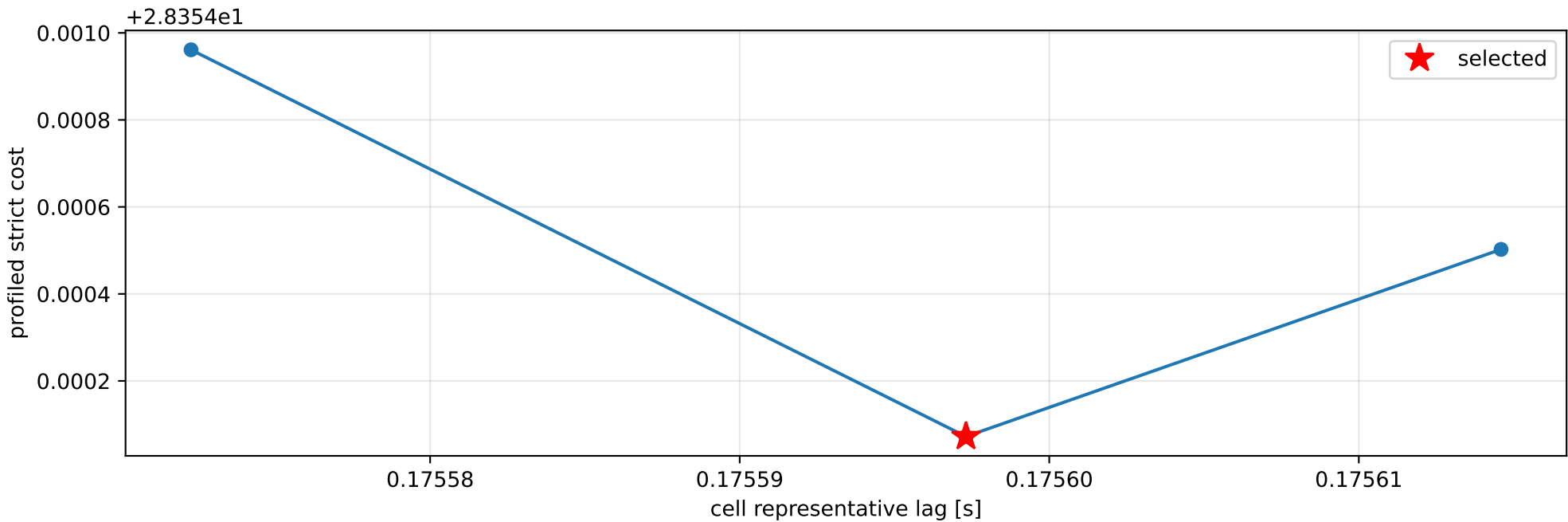
specific force [m/s^2]



inertia_over_mass_yz_nominal: smooth-to-strict continuation



inertia_over_mass_yz_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597847 s; data boundary=False

inertia_over_mass_yz_nominal: parameter estimate

Case: inertia_over_mass_yz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.89602284

inertia [kg m²]:

[[1.11897597e+00 -2.40144378e-02 8.51155360e-02]

[-2.40144378e-02 5.33109122e-01 1.23148951e-07]

[8.51155360e-02 1.23148951e-07 6.15750035e-01]]

CoG body [m]: [-0.01965494 -0.05509698 -0.00500568]

force effectiveness: [2.17554635 2.13026787 1.12590206 1.36283749]

scale-free J/m [m²]:

[[2.28547948e-01 -4.90488680e-03 1.73846281e-02]

[-4.90488680e-03 1.08886159e-01 2.51528547e-08]

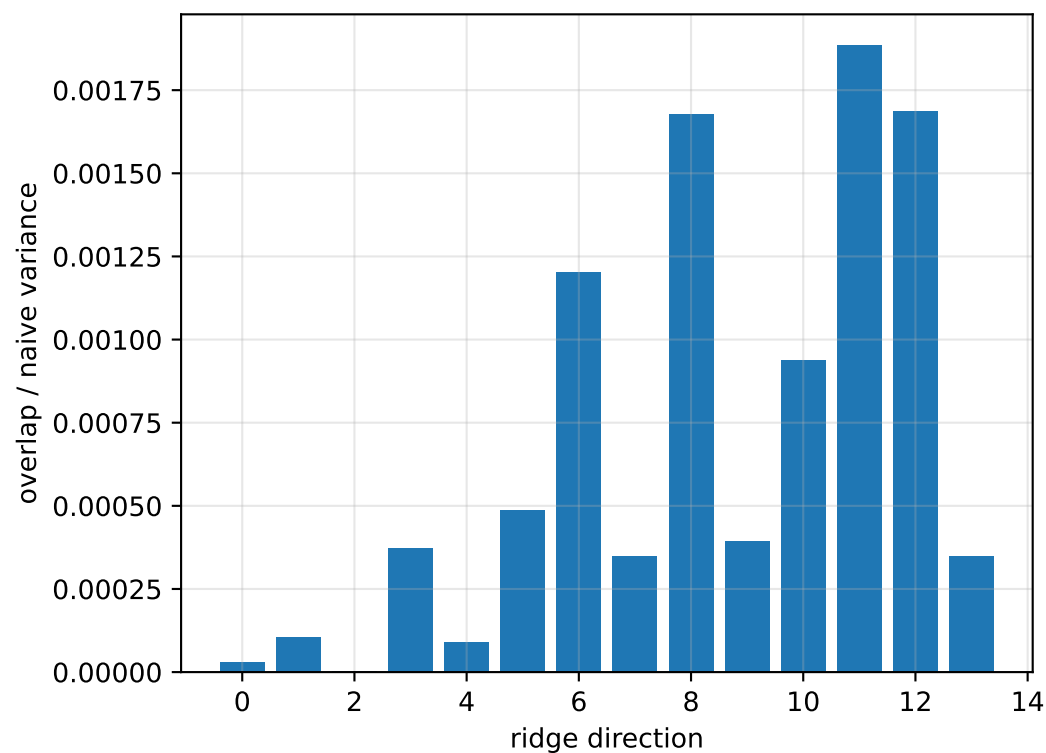
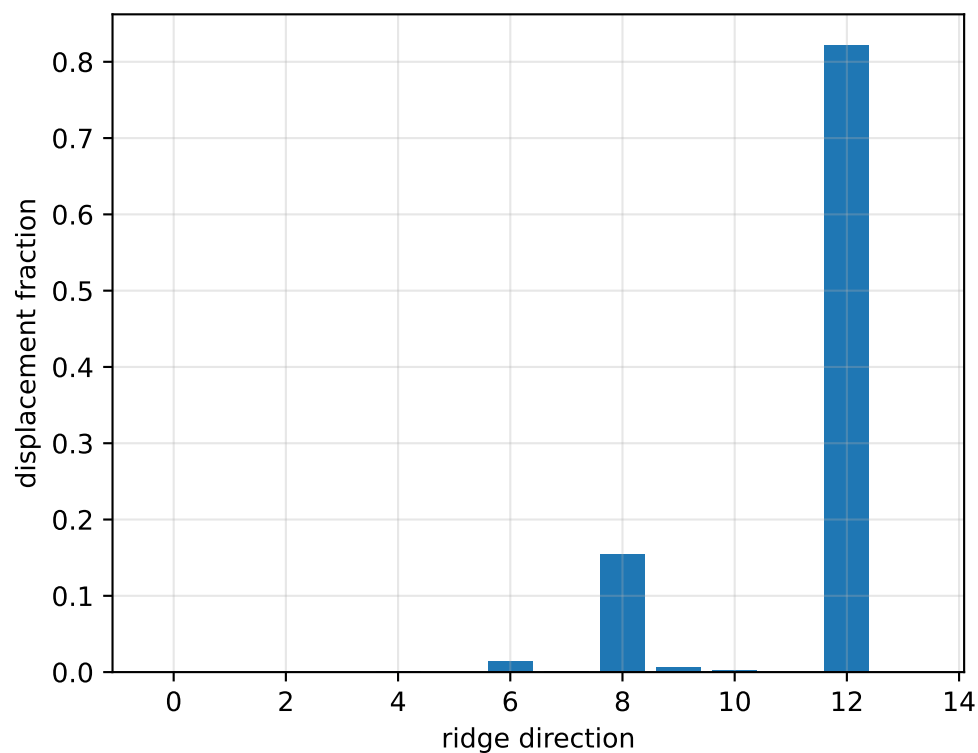
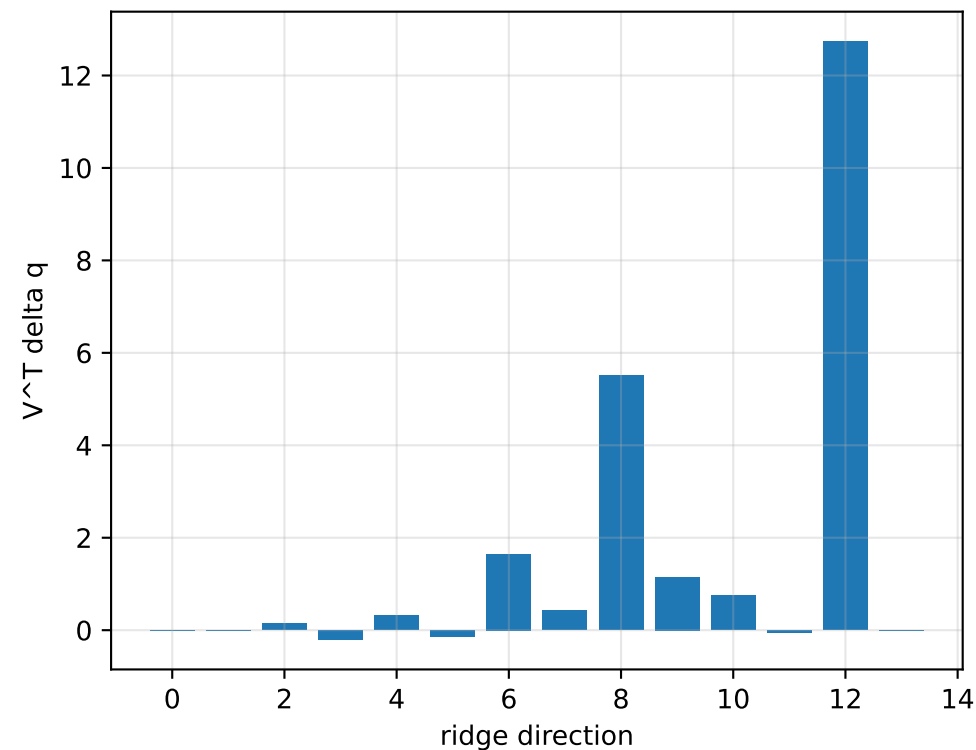
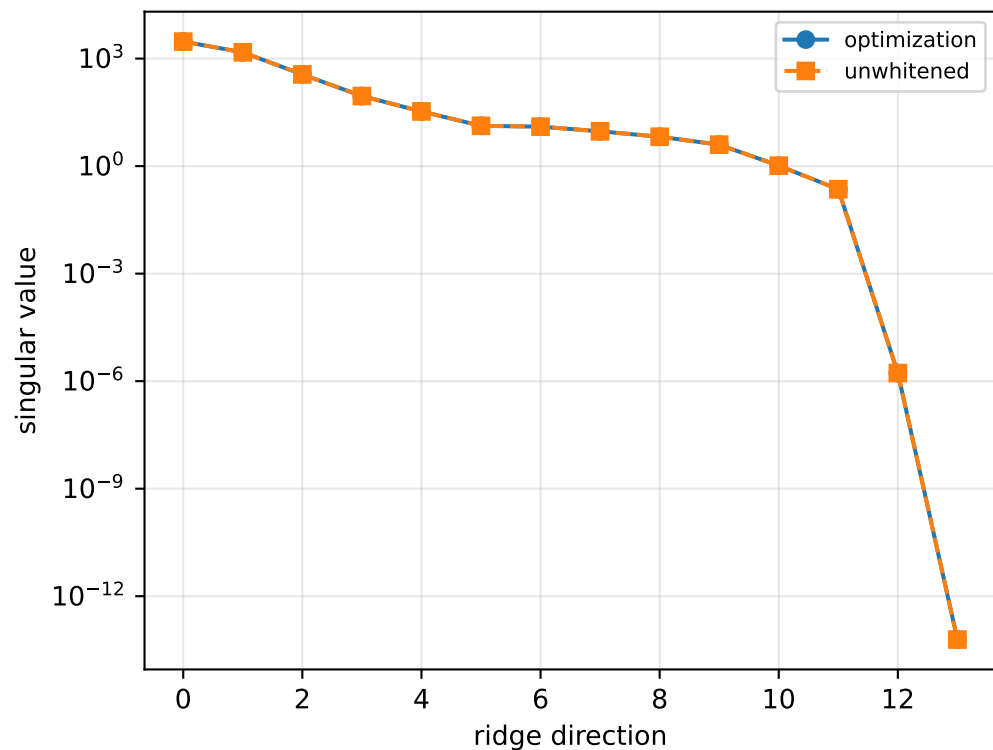
[1.73846281e-02 2.51528547e-08 1.25765352e-01]]

scale-free f/m: [0.44434971 0.4351017 0.22996258 0.27835603]

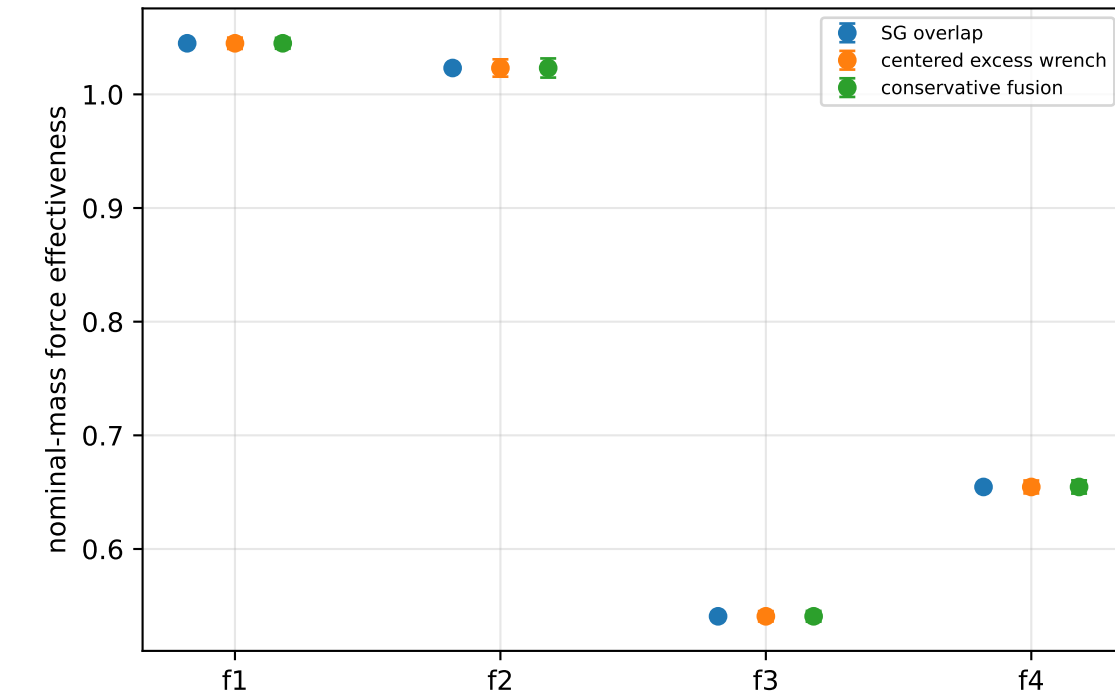
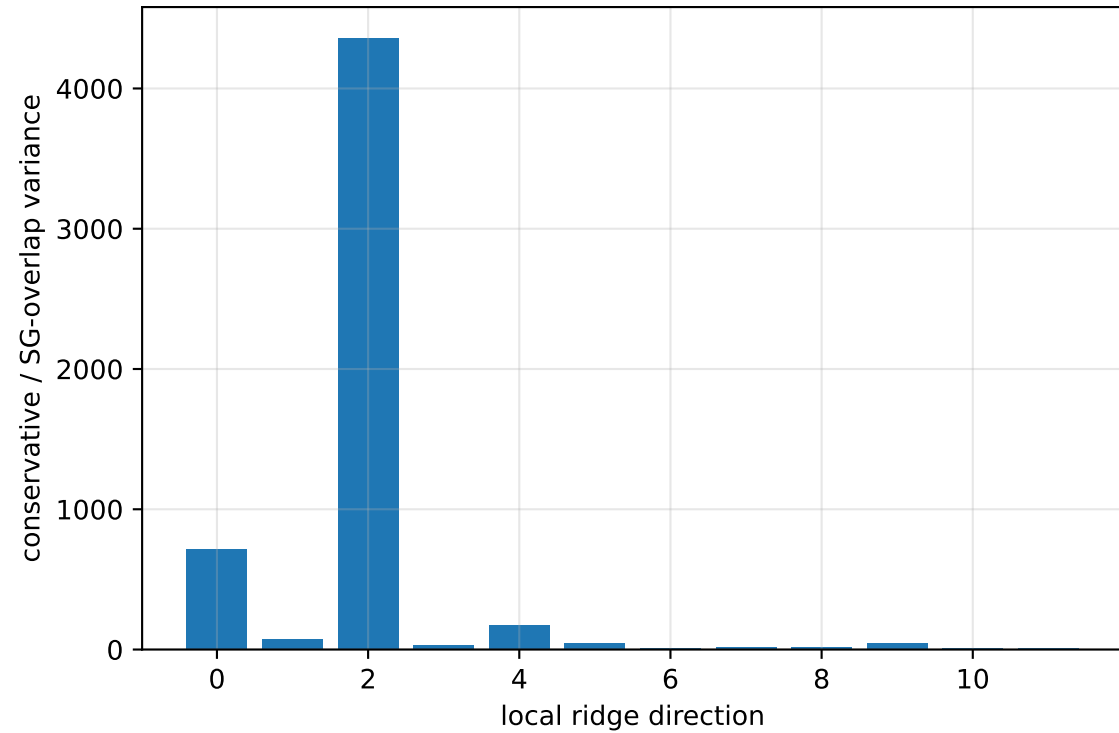
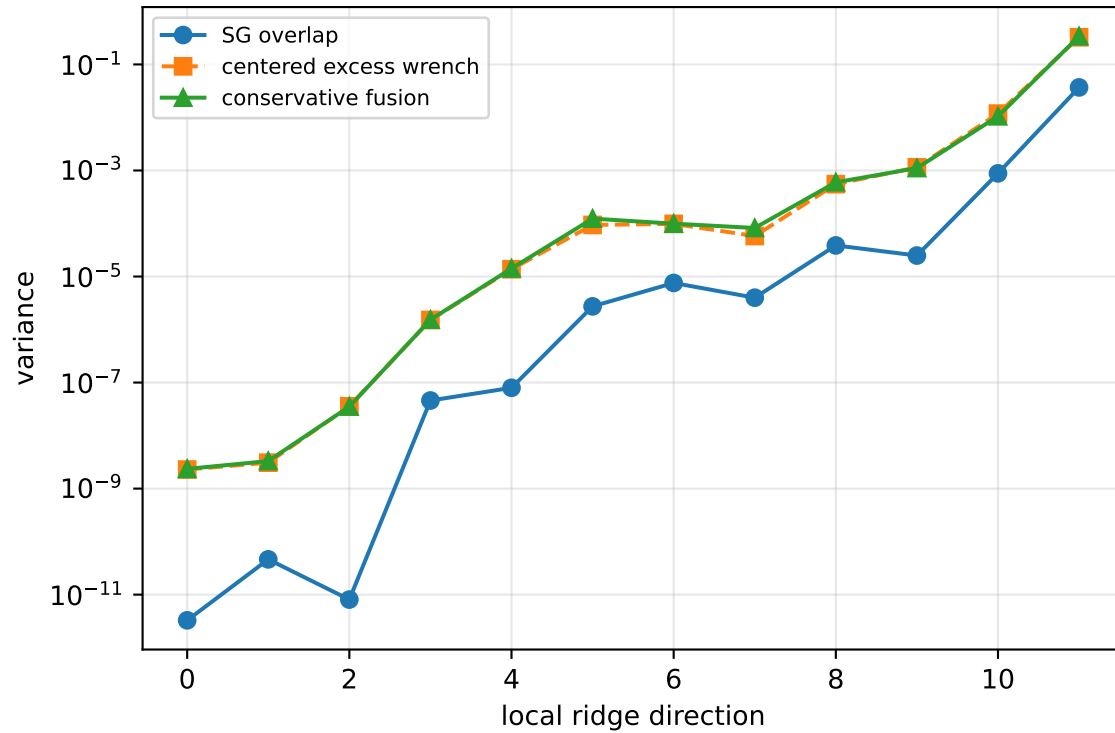
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

inertia_over_mass_yz_nominal: ridge spectrum (rank 13, nullity 1, ||v||=7.27e-14)



inertia_over_mass_yz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 411$
wrench-to-acceleration recovery max error = $3.09\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

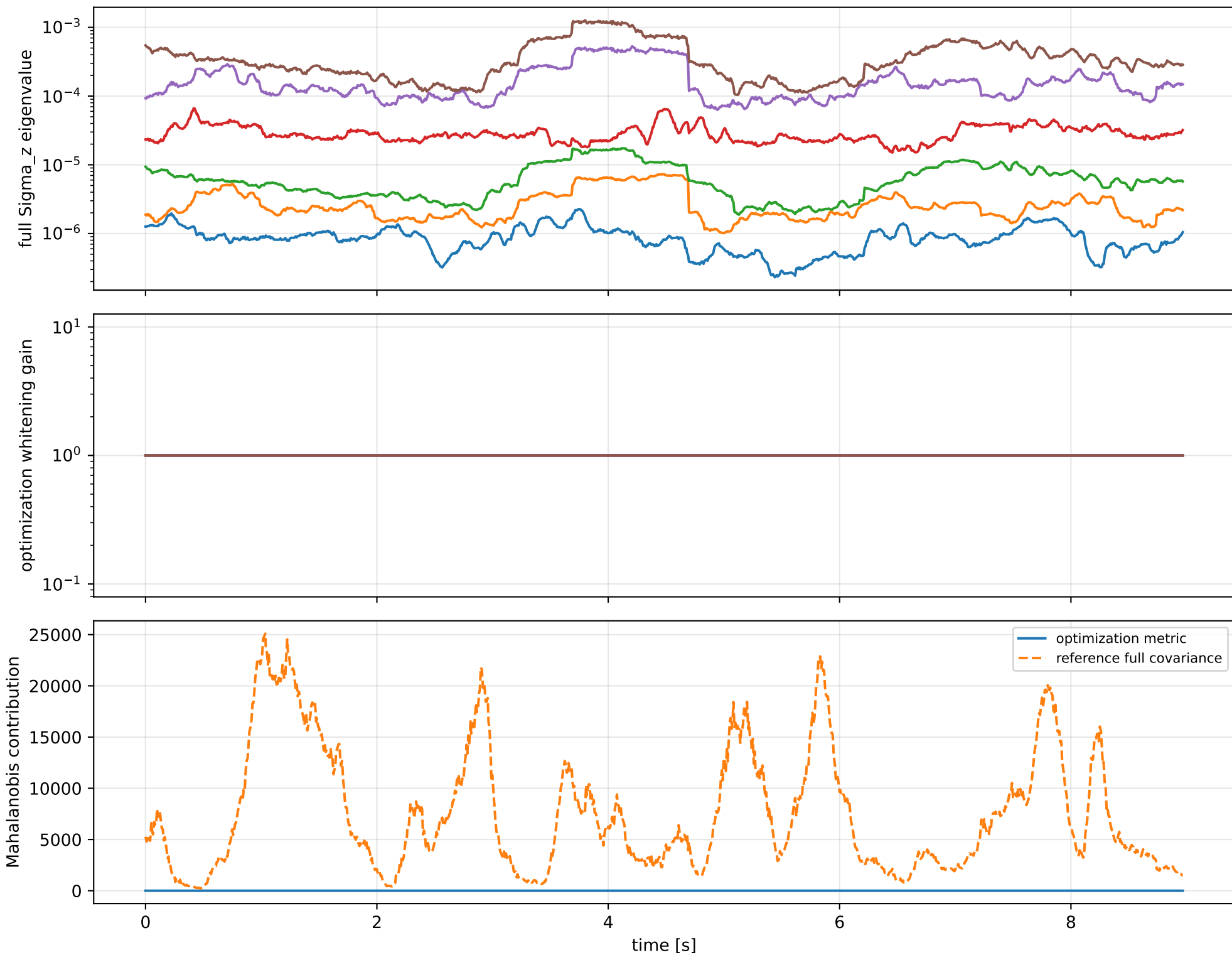
index 11: variance=0.3398, ratio=9.222
 $\log_{\text{mass}}=+0.00128$; second_moment_diag_1=+0.0577
second_moment_diag_2=-0.054; second_moment_diag_3=-0.0108
second_moment_offdiag_12=+0.994; second_moment_offdiag_13=-0.0686
second_moment_offdiag_23=+0.022; cog_x=-0.000541
cog_y=+0.000294; cog_z=-0.000431
log_force_eff_1=+0.00302; log_force_eff_2=-0.0029
log_force_eff_3=+0.00184; log_force_eff_4=+0.0039

index 10: variance=0.01051, ratio=11.91
 $\log_{\text{mass}}=-0.0196$; second_moment_diag_1=-0.0154
second_moment_diag_2=-0.0443; second_moment_diag_3=+0.148
second_moment_offdiag_12=+0.0691; second_moment_offdiag_13=+0.982
second_moment_offdiag_23=-0.0367; cog_x=+0.00638
cog_y=-0.000937; cog_z=-0.000614
log_force_eff_1=-0.0377; log_force_eff_2=+0.00744
log_force_eff_3=+0.0208; log_force_eff_4=-0.059

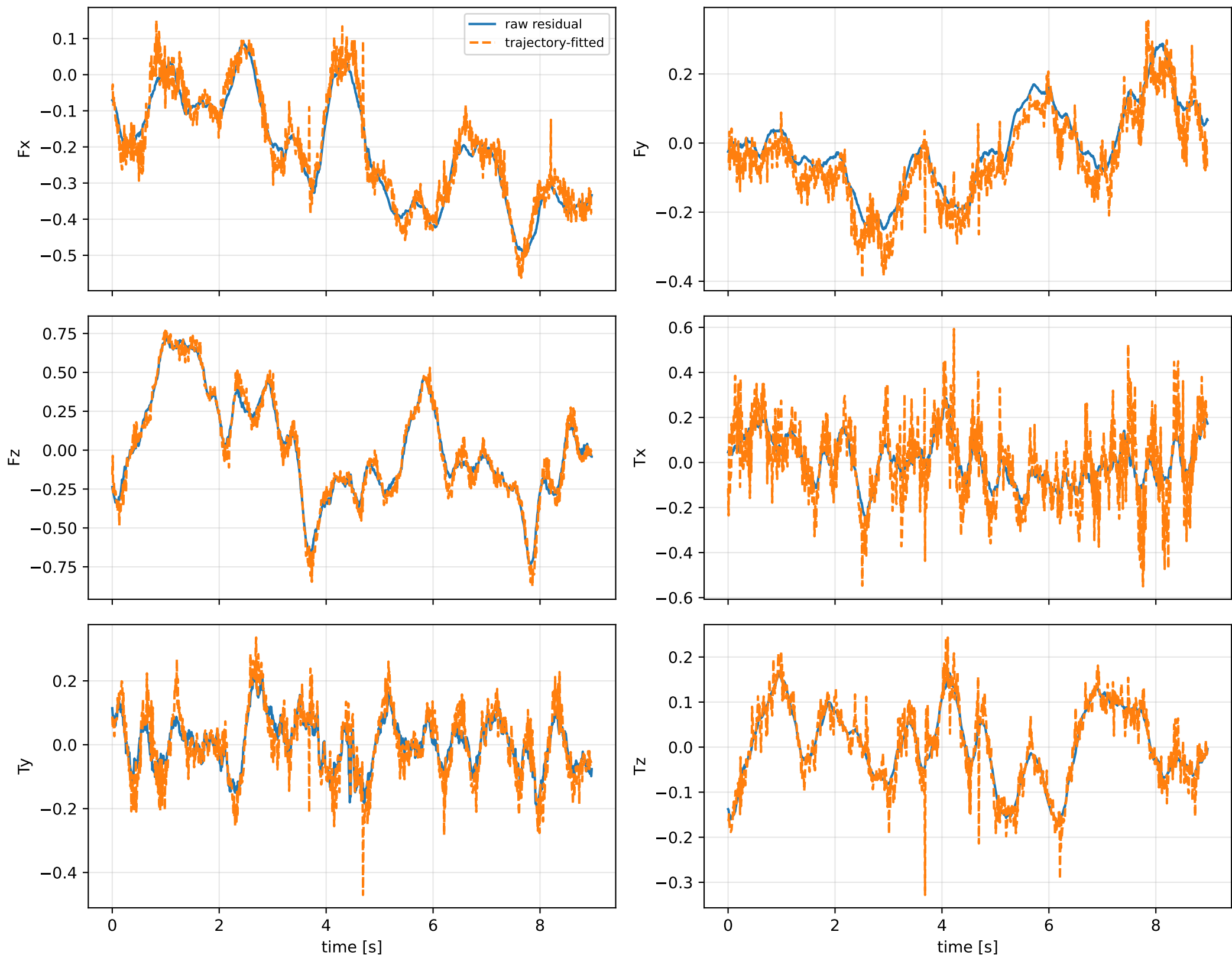
index 9: variance=0.00111, ratio=44.64
 $\log_{\text{mass}}=-0.165$; second_moment_diag_1=+0.000611
second_moment_diag_2=+0.899; second_moment_diag_3=-0.06
second_moment_offdiag_12=+0.0477; second_moment_offdiag_13=+0.0416
second_moment_offdiag_23=+0.202; cog_x=-0.00382
cog_y=-0.000761; cog_z=-0.0102
log_force_eff_1=-0.161; log_force_eff_2=-0.165
log_force_eff_3=-0.209; log_force_eff_4=-0.139

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yz_nominal: optimization vs reference covariance



inertia_over_mass_yz_nominal: wrench / closure (force max 1.52e-14, torque max 1.39e-16)



inertia_over_mass_yz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: cb359687f62672f99943f224e046f1d453f420c78c59be95963f7c1e6ba83846

data objective: 28.3540722011

prior objective: 2.90108544707e-11

total objective: 28.3540722011

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yz_nominal (inertia_over_mass_m2)

components: yz

target: [2.51604719e-08]

std: [1.e-06]

final: [2.51528547e-08]

error: [-7.61719823e-12]

standardized residual: [-7.61719823e-06]

factor objective: 2.90108544707e-11